

Extremely Low Contact Resistance on Graphene through n-Type Doping and Edge Contact Design

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Graphene has drawn scientific and technological interest for next-generation nanoelectronics beyond silicon (Si) technology because of its superior properties.^[1–4] Although graphene has been theorized to have the exceptional carrier transport property ($200\,000\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$)^[5] that is desirable in implementing radio-frequency circuits with hundred gigahertz cutoff frequencies,^[6,7] high metal–graphene (M–G) contact resistance has been hindering the successful integration of such high speed graphene devices and circuits. In the initial research stage of M–G contact resistance, metal electrodes were interfaced to the surface of 2D graphene, leading to very large contact resistance due to the lack of surface bonding sites. Many works to increase density-of-state (DOS) in graphene by using various metals were previously reported to decrease contact resistance.^[8–15] Graphene surface treatment processes by low-power O_2 and ultraviolet ozone were also used to reduce the contact

resistance (down to $200\ \Omega\ \mu\text{m}$) by enhancing the M–G interaction energy.^[16] Especially, Xia et al. thoroughly investigated a palladium (Pd)–graphene contact with a theoretical contact model based on Landauer's approach and reported a low contact resistance ($250\ \Omega\ \mu\text{m}$).^[17] Although Pd produced the low M–G contact resistance due to its n-doping effect,^[18] most works employed a strategy to maximize the Fermi-level difference (ΔE_F) between metal and graphene by selecting various metals for large graphene DOS.^[10] However, due to the difficulty in controlling the doping level in graphene, experimental approaches to change the Fermi-level of graphene through doping technique have not yet been reported for the accomplishment of low contact resistance.

Meanwhile, in the recently reported M–G contact studies, new contact concept where metal electrodes are faced to 2D graphene along 1D graphene edge or cut-side with larger DOS has been investigated to minimize the M–G contact resistance.^[19–21] Wang et al. achieved very low contact resistance of $150\ \Omega\ \mu\text{m}$ on monolayer exfoliated graphene through a 1D edge contact method, exhibiting superior mobility of $140\,000\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ at $2 \times 10^{11}\text{ cm}^{-2}$.^[22] Leong et al. reported another graphene edge contact method by metal-catalyzed etching in hydrogen, resulting in a contact resistance of $100\ \Omega\ \mu\text{m}$ (on monolayer graphene flake), which is the lowest contact resistance reported so far.^[19] However, in this case of edge contact, there also has been no attempt to reduce the M–G contact resistance by doping graphene (thereby, increasing DOS). In addition, there is a lack of quantitative analysis of the edge contact scheme through variously designed patterns. Here, we investigate the effects of graphene n-doping on the M–G contact in combination with edge contact, presenting a record M–G contact resistance on graphene grown by a chemical vapor deposition (CVD) technique. The n-doping is achieved by the negative charge transfer phenomenon at the interface between graphene and poly(4-vinylphenol)/poly(melamine-co-formaldehyde) (PVP/PMF) insulator. Besides, the currents through the 2D surface and edge are experimentally investigated with the designed patterns. Finally, we have applied this contact scheme to graphene–perovskite hybrid photodetector for the performance improvement and also minimized the contact resistance to $23\ \Omega\ \mu\text{m}$, which is the lowest value reported to the best of our knowledge and also lower than the required value in the latest Si CMOS technology.

Traditional doping process by ion implantation on silicon (Si) was not effective on doping a single graphene layer because the ion collisions introduced too much damage to the surface,

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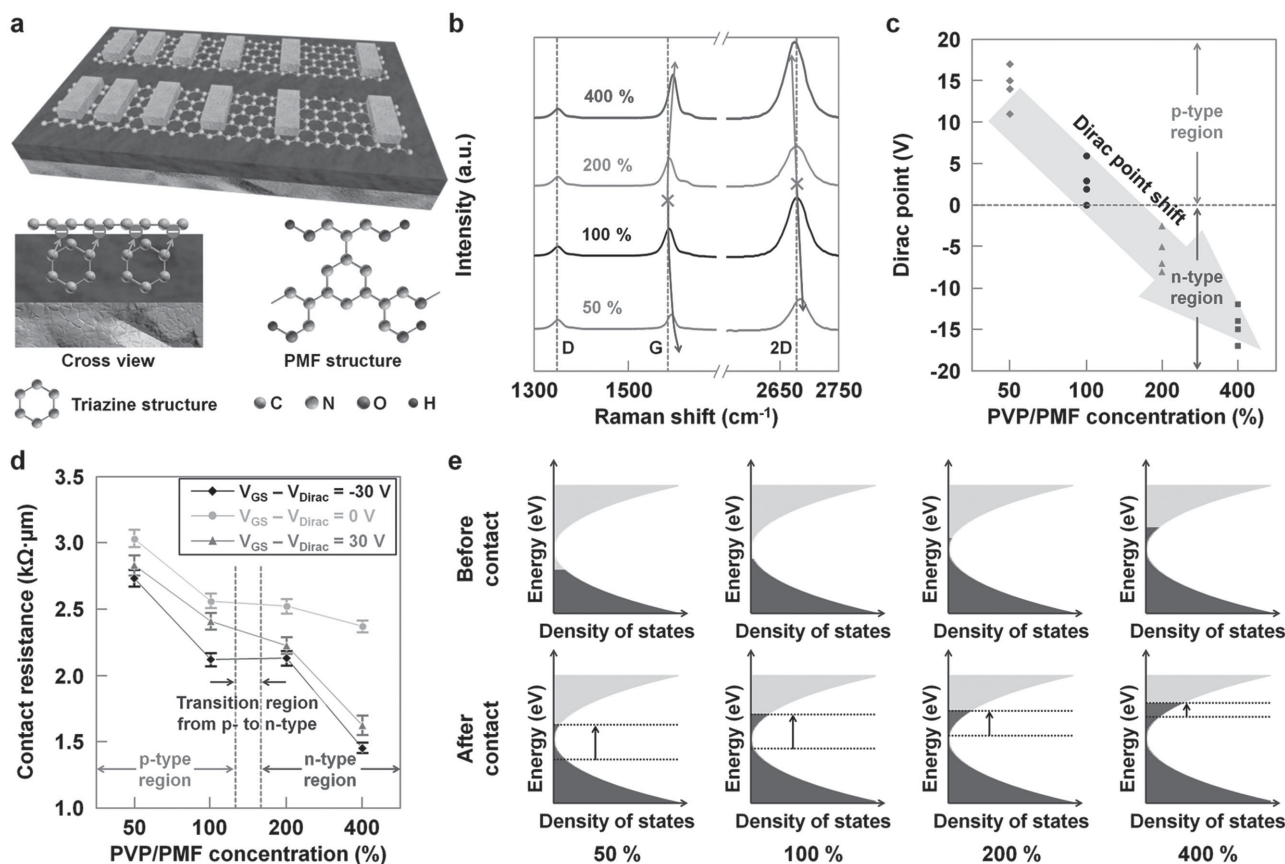


Figure 1. a) Schematic diagrams explaining the graphene doping mechanism by PVP/PMF layer. b) Raman spectra of graphene doped by PVP/PMF layers with various concentrations (50%, 100%, 200%, and 400%). c) Distribution of Dirac points extracted from the I_D - V_G characteristics for each PVP/PMF concentration. d) Contact resistance as a function of PVP/PMF concentration in various vertical electric fields ($V_{GS} - V_{Dirac} = -30$ V, 0 V, and +30 V). e) Schematic diagrams of charge (electron) transfer phenomenon from Ti.

eventually destroying the delicate layer. Doping by PVP/PMF is more preferred to graphene for it is nondestructive and does not require very high temperature annealing for dopant activation. As shown in Figure 1a, triazine functional groups in PMF are electron-rich aromatic molecules because nitrogen atoms in the triazine groups work as electron donors.^[23] When this layer comes in contact with graphene, negative charges (electrons) are expected to transfer from triazine functional groups to graphene. This effectively acts as an n-doping source, making it feasible for graphene based device fabrication. For graphene device to be practical to the industry, one must be able to fully control the degree of doping level (thus, the Fermi-level) in graphene. In order to control the graphene Fermi-level, we adjust the amount of PMF molecules in PVP/PMF insulating layer, which is in contact with the graphene. The fabrication process based on an inverse graphene transfer method using PVP/PMF layer is explained in Figure S1 in the Supporting Information. Since Raman spectroscopy is a widely known method of observing the type of doping^[24] in graphene, by observing the relative peak shift of the doped graphene to the as-grown sample, one can distinguish whether the graphene is n-type or p-type. Figure 1b shows Raman spectra measured on graphene layers, which are transferred onto the PVP/PMF insulating layers with different PMF concentrations (50–400%). Here,

the reference peak position of intrinsic graphene is indicated by cross marks. We first observed the redshift of G peaks in graphene with 50% and 100% PVP/PMF layers when compared to that of as-grown graphene, which is expected to be strong p-type due to native hydroxyl (–OH) functional groups under atmospheric ambient. Then, the G peaks blueshifted as the PMF concentration was increased beyond 200% PMF to 400%, indicating that the doping type of graphene has changed from p-type to n-type. Based on the previously reported Raman analysis on p- and n-doped graphene,^[24] intrinsic graphene is expected to be achieved between 100% and 200% PVP/PMF concentrations. The 2D peaks of the graphene on PVP/PMF layers were redshifted as the PMF concentration increased because the electron concentration was further increased by the negative charge (electron) transfer phenomenon at the graphene–PVP/PMF interface. The brightness of Raman mapping images in all cases of PVP/PMF concentrations was uniformed (Supporting Information, Figure S2), indicating that the PVP/PMF insulating layers uniformly doped the graphene layers. Similar Raman G and 2D peak shifting behavior according to the doping process was previously reported by Sood and co-workers and other numerous researching groups.^[24–26] We also performed additional Raman analysis on the graphene sample after being as-grown on Cu foil, after being transferred onto

PVP/PMF 400% layer, and after being n-doped by PVP/PMF 400% layer to monitor the D-to-G peak ratio change during each fabrication step (refer to Figure S3 in the Supporting Information). Here, very similar average D-to-G peak ratio values (0.495–0.497) with low standard deviation values (0.013–0.016) were obtained after each step. This indicates that the graphene layer was not affected much by the process damage and contamination during the fabrication. In particular, compared to the D-to-G peak ratio values of a high quality graphene (about 0.1), the higher values in this work suggest that the graphene layer used in this work is relatively poor in crystal quality due to small crystal grain size. The Dirac points were extracted from I_D - V_G curves (refer to Figure S5 in the Supporting Information) of graphene transistors with four different PVP/PMF films, also coinciding with the suggested n-type doping effect by PMF molecules in Figure 1c. As the PMF concentration increases, the Dirac point shifts toward negative voltage region, which indicates an increase in electron concentration. In particular, PVP/PMF layer with above 200% PMF concentration turns the graphene into n-type ($V_{\text{Dirac}} = -5$ V for 200% PVP/PMF and $V_{\text{Dirac}} = -13$ V for 400% PVP/PMF), but p-type property still remained when lower concentration PVP/PMF layers were used ($V_{\text{Dirac}} = 14$ V for 50% PVP/PMF and $V_{\text{Dirac}} = 3$ V for 100% PVP/PMF). Additionally, we have performed UV photoelectron spectroscopy (UPS) analysis to confirm the difference in the graphene workfunctions on the PVP/PMF layers (PVP/PMF 50%, 100%, 200%, and 400%) (see Supporting Information, Figure S7).

We then investigated the M-G contact resistances (Figure 1d) on the four kinds of graphene layers by using a transfer length method (TLM) structure using Ti as a contact metal (refer to Supporting Information, Figure S5). The contact resistance values under various vertical electric fields ($V_{\text{GS}} - V_{\text{Dirac}} = -30, 0, \text{ and } +30$ V) were plotted with error bars as a function of PVP/PMF concentration (each data point is composed of four TLM measurements). Here, the M-G contact measurement was performed four times with four kinds of TLM sets in each graphene doping condition. Although slightly lower contact resistances were observed under high negative gate bias, a similar dependence on PVP/PMF concentration (decreasing trend) was confirmed. In the PVP/PMF 50% sample, relatively higher contact resistance was obtained ($R_C = 2732 \Omega \mu\text{m}$ at $V_{\text{GS}} - V_{\text{Dirac}} = -30$ V) compared to samples with different concentration. As shown in Figure 1e, it is predicted that charge (electron) transfer phenomenon from Ti (the calculated workfunction after the formation of M-G contact is 4.17 eV)^[18] to the p-type graphene on 50% PVP/PMF makes the graphene to be close to intrinsic, subsequently reducing DOS and increasing contact resistance. However, in higher PMF concentration samples (PVP/PMF 100% and 200%), the contact resistance was slightly decreased to $2100 \Omega \mu\text{m}$ at $V_{\text{GS}} - V_{\text{Dirac}} = -30$ V. This is due to electron transfer from Ti that additionally increases the Fermi-level toward a higher DOS region (purple region) when graphene Fermi-level moves up to near intrinsic level by n-type doping effect (gray region in Figure 1e). After doping graphene with 400% PVP/PMF layer, much lower contact resistance of $1454 \Omega \mu\text{m}$ was obtained under the same gate bias. In the case of doped graphene by 50% PVP/PMF layer, even though its sheet resistance was lower than those of graphene

on 100% and 200% PVP/PMF layers, the current level was the lowest because it exhibited the highest contact resistance (refer to Supporting Information, Figure S5). On the other hand, the highest current was observed in the graphene on 400% PVP/PMF film, since both contact resistance and sheet resistance were lower than the other samples.

In order to investigate the dependence of contact resistance on metal type before and after the 400% PVP/PMF doping, we repeated the experiment using Pd and Cu as contact metals, which induce different chemisorption of graphene after the M-G contact formation. Figure 2a,d shows the total resistance versus channel length from which the contact and sheet resistances in the devices with various contact metals are extracted when V_{GS} of $V_{\text{Dirac}} - 30$ V is applied. The graphene devices were measured before and after the 400% PVP/PMF doping because the graphene doping effect could be achieved on the PVP/PMF layer by a simple 180 °C anneal for 30 min in a vacuum oven. As shown in Figure 2b, on the undoped graphene layer which shows p-type property, Cu-graphene contact presented much higher contact resistance ($2110 \Omega \mu\text{m}$) than Ti and Pd (1890 and $1533 \Omega \mu\text{m}$, respectively). When compared to Cu-graphene with equilibrium separations (d_{eq}) of $3.26 \text{ \AA}$ and binding energy of 0.033 eV ,^[18] it is thought that the smaller equilibrium separations ($d_{\text{eq}} = 2.10$ and $2.30 \text{ \AA}$, respectively) and the higher binding energy values per carbon atom ($\Delta E = 0.327$ and 0.084 eV , respectively) at the Ti and Pd-graphene contacts induce substantial chemisorption on graphene. This stronger chemisorption transfers more number of electrons from metal to graphene (n-type doping of graphene), consequently achieving higher DOS and lower contact resistance. Although graphene is known to be also n-type doped by Cu, this n-doping phenomenon is much weaker because graphene is physisorbed on the Cu.^[18] In Figure 2c that schematically illustrates the M-G interfaces on undoped graphene, the largest and smallest increases in Fermi-level (indicated by purple in the DOS functions) are, respectively, expected in Pd and Cu contact samples, regardless of their workfunctions (5.22 eV for Cu, 4.56 eV for Ti, and 5.67 eV for Pd). Khomyakov et al.^[18] previously calculated the workfunctions of the graphene layers absorbed chemically or physically on various metals, where the workfunctions of graphene in Cu, Ti, and Pd-graphene contacts were, respectively, 4.40 , 4.17 , and 4.03 eV . This workfunction difference was also confirmed by UPS analysis (see Supporting Information, Figure S7). In particular, as already known by many reports,^[17,18] the contact resistance in Pd-graphene contact was lower than that of Ti-graphene despite of its longer equilibrium separation and lower binding energy. This seems to be attributed to the weaker charge transfer phenomenon by Ti than Pd, which leads to a smaller DOS increase in the graphene under the Ti.^[18] Xia et al. reported a similar result on exfoliated monolayer graphene flake that the contact resistance ($470 \Omega \mu\text{m}$ at room temperature (RT)) at Ti-graphene interface is two times as larger as that ($248 \Omega \mu\text{m}$ at RT) of the Pd-graphene. However, the difference in contact resistances according to metal type was not significant after strongly n-doping the graphene with PVP/PMF 400% layer in Figure 2e. The Cu, Ti, and Pd-graphene contacts were respectively 1539 , 1454 , and $1381 \Omega \mu\text{m}$ at $V_{\text{GS}} - V_{\text{Dirac}} = -30$ V, which were also lower than that of undoped Pd-graphene samples. It is thought that the n-doping moves the graphene Fermi-level up to the lower energy level where the DOS

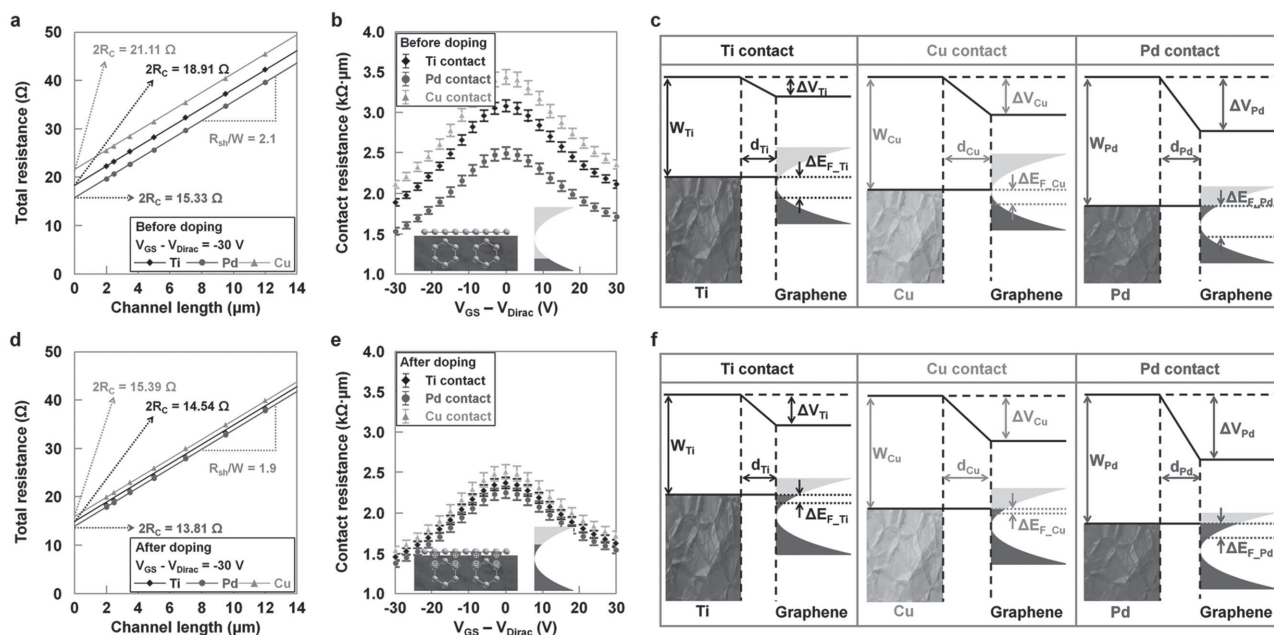


Figure 2. a,d) Total resistance versus channel length plots to extract contact and sheet resistances for various contact metals (Ti, Cu, Pd) before (a) and after (d) 400% PVP/PMF doping at $V_{GS} - V_{Dirac} = -30$ V. b,e) Extracted contact resistance as a function of $V_{GS} - V_{Dirac}$ in various metals before (b) and after (e) doping. c,f) Schematic illustrations of metal and graphene junction interfaces in un-doped (c) and doped (f) graphene for each contact metal.

is relatively high, as seen in Figure 2f, lowering the contact resistance. Although the similar electron injection from the metals is also expected after the n-doping process, it does not affect the graphene Fermi-level in a similar manner compared to previous undoped graphene samples due to the larger DOS (indicated by the purple area in Figure 2f), also making the contact resistance to be less dependent on the metal type.

Next, we experimentally investigated 2D surface and edge contacts on graphene by using various patterns with different perimeter-to-area ratios under a Pd electrode, which previously showed the lowest contact resistance value. Here, the three different patterns shown in Figure 3a were used to form an edge contact on graphene, and the distance between the edge patterns and the metal edge (L_{EM}) was fixed as 15 μm to verify the surface current in a clear manner. Edge 2 and 3 patterns, respectively, have 5 and 1 μm wide periodic lines/gaps. For edge patterns, there are two different current paths through surface and edge regions, which are, respectively, defined by pink and cyan dotted lines. In the case of surface current, its effective transfer length is known to be about 1 μm ($L_{T,2D}$), making it possible to assume that the amount of surface currents is almost same in all of the samples regardless of the edge patterns. Edge 3 pattern is expected to show the highest edge current because of dramatic perimeter length increases in effective transfer length ($L_{T,1D}$). We then extracted edge currents from the three edge contact patterns by excluding the surface current components (Supporting Information, Figure S9), and the edge currents were plotted as a function of edge pattern in Figure 3b. The currents are also shown as a function of $V_{GS} - V_{Dirac}$ in Supporting Information, Figure S9b, comparing it with the surface current. Considering that the current level in the edge 3 pattern is roughly three times higher than that of the edge 2 (also, $I_{Edge2}/I_{Edge1} = 2$), the effective transfer length for the edge

current ($L_{T,1D}$) is estimated to be about 5 μm in the patterns with 15 μm surface contact region. In addition, we predict that the edge contact is more dominant current path than the surface contact, because the edge currents showed a higher value than surface currents in edge 2 and 3 patterns despite the separation by a long spacing of 15 μm between edge patterns and the metal edge (Supporting Information, Figure S9). Here, relatively lower electric field is expected for edge current when compared to the surface current. As a result, compared to the surface contact sample ($R_C = 1.4$ kΩ μm at $V_{GS} - V_{Dirac} = -30$ V), we obtained about three times lower contact resistance ($R_C = 484$ Ω μm at $V_{GS} - V_{Dirac} = -30$ V) in the edge 3 pattern.

Figure 4a shows the schematic diagram of surface and edge contacts by Ti, Cu, and Pd to graphene presenting the effect of different metal type on contact resistance in surface and edge contacts. For the comparison of the contact resistances among different metal contacts, TLM analysis is conducted using edge 3 pattern and the resulting TLM plot (total resistance versus channel length) is shown in Figure 4b. Here, Ti, Cu, and Pd plots show the same slopes while Ti has an overall lower total resistance. This signifies that while Ti, Cu, and Pd contacts have the same sheet resistance, only the edge contact resistance is different. For edge contacts, Ti (83 Ω μm) contact exhibited lower contact resistance than Cu and Pd (254 Ω μm for Cu and 484 Ω μm for Pd). It is thought that the higher equilibrium interaction energy values per carbon atom ($E_{Ti} = 77.4$ kcal mol⁻¹) and higher projected density of states (PDOS) per unit cell of the metal atoms (PDOS(Ti) = 15 eV⁻¹) at the Ti contact induced a smaller contact resistance, compared to the equilibrium interaction energy and PDOS values of Cu and Pd ($E_{Cu} = 45.2$ kcal mol⁻¹, $E_{Pd} = 55.9$ kcal mol⁻¹, PDOS(Cu) = 5 eV⁻¹, and PDOS(Pd) = 6 eV⁻¹).^[27] Figure 4d shows the contact resistance values in Ti, Cu, and Pd edge contacted graphene

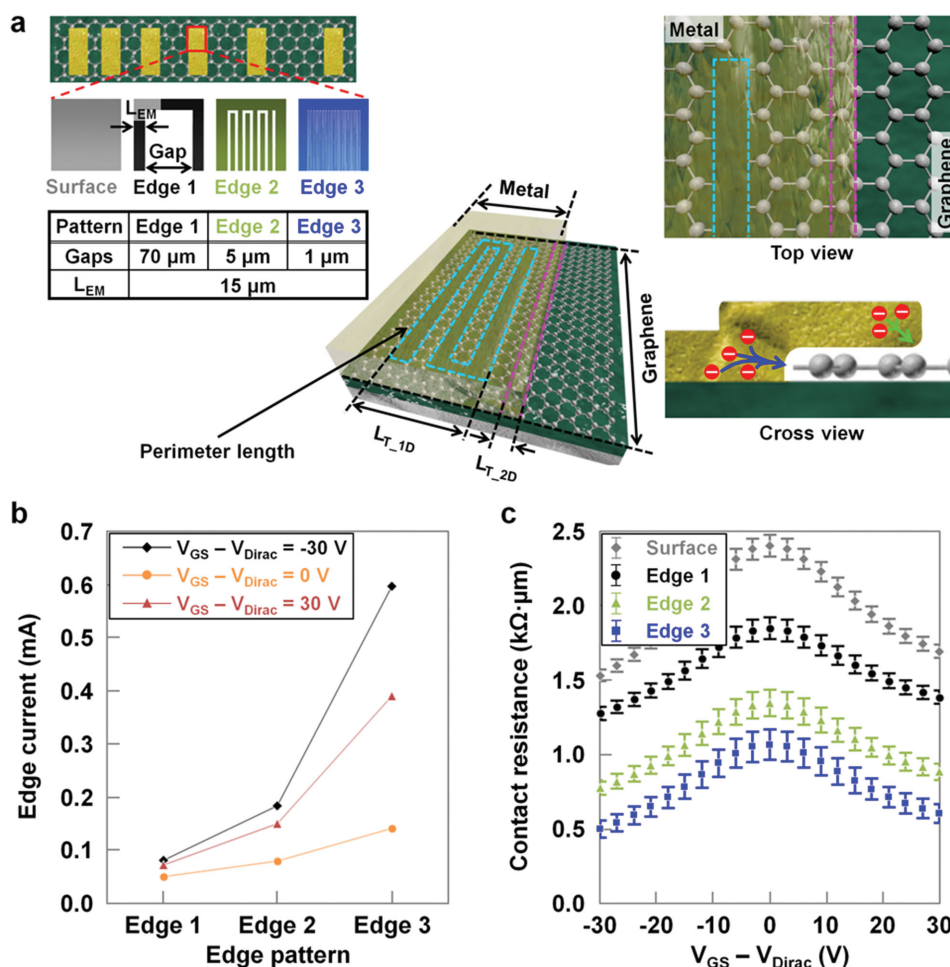


Figure 3. a) Schematic diagrams of different contact patterns (surface, edge 1, edge 2, and edge 3). b) Edge currents according to edge pattern in various vertical electric fields ($V_{\text{GS}} - V_{\text{Dirac}} = -30$ V, $V_{\text{GS}} - V_{\text{Dirac}} = 0$ V, and $V_{\text{GS}} - V_{\text{Dirac}} = +30$ V). c) Extracted contact resistance as a function of $V_{\text{GS}} - V_{\text{Dirac}}$ in different contact patterns (surface, edge 1, edge 2, and edge 3).

before and after doping. Here, the graphene is doped with PVP/PMF 400% and edge 3 patterns were used in the contact scheme. Comparing the contact resistances in the edge contacted graphene before and after the doping, it is found that contact resistance has decreased in accordance with the metal's workfunction ($R_{\text{C}} = 42 \Omega \mu\text{m}$ for Ti, $R_{\text{C}} = 92 \Omega \mu\text{m}$ for Cu, and $R_{\text{C}} = 122 \Omega \mu\text{m}$ for Pd) which is different from the case of surface contacts. Next, to confirm the effect of contact resistance reduction by doping and edge contact in terms of device performance, we fabricated a perovskite/graphene photodetectors with two kinds of patterns (surface and edge 3), where the perovskite layer was employed to improve the absorbing characteristics of graphene. Figure 4e shows the schematic diagram of the perovskite/graphene photodetector under illumination. The current–voltage ($I_{\text{D}} - V_{\text{G}}$) curves measured under dark and illuminated condition ($P_{\text{Light}} = 1$ mW, $\lambda = 520, 655, 785$, and 850 nm) can be found in Figure S10 in the Supporting Information. To investigate the performance improvement obtained by applying the edge contact scheme, we extracted and plotted the photoresponsivity ($R = I_{\text{Photo}}/P_{\text{Light}}$) and detectivity ($D^* = (RA^{1/2})/(2eI_{\text{Dark}})^{1/2}$) values as a function of wavelengths from

the $I_{\text{D}} - V_{\text{G}}$ curves, as seen in Figure 4f,g. Here, the photocurrent is $I_{\text{Laser_on}} - I_{\text{Laser_off}}$, P_{Light} is the total incident optical power, A is the effective detection area of the device, e is the absolute electron charge value (1.6×10^{-19} C), and I_{Dark} is the dark current. In the case where a 520 nm laser was exposed, the photoresponsivity and detectivity were enhanced about three times in the edge contacted perovskite/graphene photodetector, compared to the surface contacted device. It is thought that the higher electrical field achieved by the contact resistance reduction at the Ti–graphene edge contact improves the ability of carrier collection. Although the photoresponsivity and detectivity decreased as the wavelength increased (due to perovskite absorption range limit), the photoresponsivity and detectivity in the edge contacted photodetector are still higher than those in the surface contacted photodetector. Finally, in order to minimize further graphene contact resistance by combining n-doping and edge contact schemes, edge pattern was changed; the 1 μm spacing was reduced to 150 nm and the edge patterns with 150 nm spacing were added onto the surface contact area (edge pattern 4). As a result, we obtained a very low contact resistance ($23 \Omega \mu\text{m}$) at RT due to perimeter length increase,

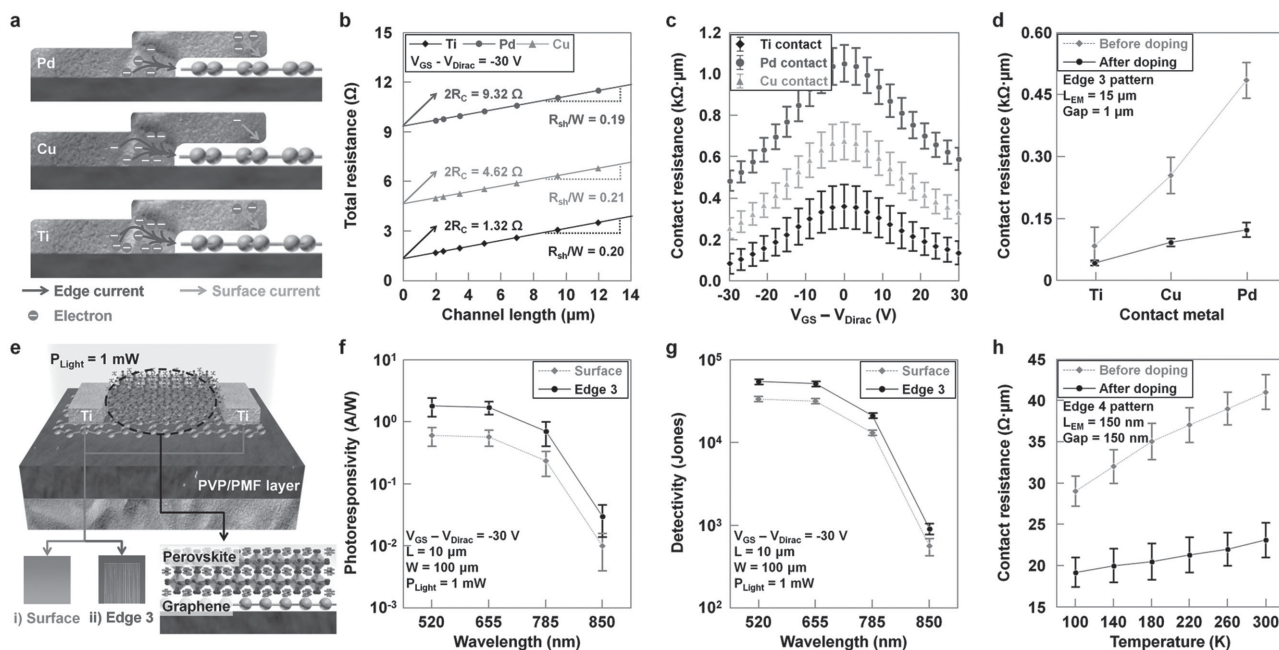


Figure 4. a) Schematic diagrams of surface and edge contacts by Ti, Cu, and Pd to graphene presenting the effect of different metal type on contact resistance in surface and edge contacts. b) Total resistance versus channel length plots to extract the contact and sheet resistances, which were obtained in edge 3 patterned samples with various contact metals (Ti, Cu, and Pd). c) Extracted contact resistance as a function of $V_{GS} - V_{Dirac}$ in different contact metals (Ti, Cu, and Pd) before doping. d) Contact resistance values as a function of different metals before and after 400% PVP/PMF doping. e) Schematic diagram of graphene/perovskite photodetector. f) Photoresponsivity and g) detectivity as a function of edge patterns under exposure of lasers with various wavelengths (405, 520, 785, 850, and 980 nm). h) Extracted contact resistance as a function of temperature in edge 4 patterned samples before and after 400% PVP/PMF doping.

which is two times lower than the contact resistance of edge pattern 3 ($41 \Omega \mu\text{m}$ at RT). Here, a V_{GS} of ($V_{Dirac} - 30$) V was applied and Ti electrode was used to form the contact. In addition, very low contact resistance of $19 \Omega \mu\text{m}$ was observed when the temperature was cooled down to 100 K. The reason behind the contact resistance reduction with temperature originates from increase in mean free path and mobility in graphene due to a temperature decrease.^[22]

In summary, we investigated the effects of graphene n-doping on M–G contact resistance in combination with edge contacts, presenting a record contact resistance of $23 \Omega \mu\text{m}$ at room temperature ($19 \Omega \mu\text{m}$ at 100 K). The graphene n-doping was achieved through the charge transfer from PVP/PMF insulator with triazine functional groups which are electron-rich aromatic molecules. The degree of doping was also controlled by adjusting the ratio of PMF to PVP. Through n-doping the graphene by 400% PVP/PMF layer, about two times lower contact resistance value was obtained compared to the 2D surface contact resistance on pristine graphene (initially, p-type) under the same gate bias. However, after the n-doping, the 2D contacted metals did not affect the Fermi-level of graphene similarly to the previous pristine graphene samples due to the already achieved larger DOS, making the contact resistance to be less dependent on the metal type. In addition, to understand the impact of edge contact on the M–G contact resistance, currents flowing through the 2D graphene surface and patterned graphene edge were experimentally investigated by using various patterns with different perimeter-to-area ratio under contact metal. As a result, compared to the 2D contact sample

($R_C = 1.4 \text{ k}\Omega \mu\text{m}$), we confirmed about three times lower contact resistance ($R_C = 484 \Omega \mu\text{m}$) in the edge 3 pattern with $1 \mu\text{m}$ wide periodic lines/gaps. In particular, for edge contacts, Ti ($83 \Omega \mu\text{m}$) contact exhibited lower contact resistance than Cu and Pd ($254 \Omega \mu\text{m}$ for Cu and $484 \Omega \mu\text{m}$ for Pd). This contact scheme was applied to the graphene–perovskite hybrid photodetector, significantly improving its performance ($0.6 \rightarrow 1.8 \text{ A W}^{-1}$ in photoresponsivity and $3.3 \times 10^4 \rightarrow 5.4 \times 10^4$ Jones in detectivity). Furthermore, to minimize further graphene contact resistance by combining n-doping and edge contact schemes, $1 \mu\text{m}$ spacing in edge pattern 3 was reduced to 150 nm and the edge patterns with 150 nm spacing were added onto the surface contact area (edge pattern 4), eventually resulting in a very low contact resistance of $23 \Omega \mu\text{m}$. We also note that, compared to the 2D contact device before doping, we obtained about 3.3 times higher field-effect mobility in the edge 4 patterned sample after doping (see Supporting Information, Figure S11).

Experimental Section

Preparation of the PVP/PMF Solution: The 0.5 g of PVP was mixed with 4.25, 4, 3.5, and 2.5 g of propylene glycol methyl ether acetate to produce PVP solution. Then, different PMF amount (0.25, 0.5, 1, and 2 g) was added to the PVP solution, yielding different PVP/PMF concentration (50%, 100%, 200%, and 400%).

Graphene Transfer Process by PVP/PMF Layer: The graphene layer was synthesized on copper foil by the conventional CVD method. PVP/PMF was spin-coated onto the graphene to form PVP/PMF dielectric layer for doping graphene. The Cu foil was etched in ammonium persulfate

$(\text{NH}_4)_2\text{S}_2\text{O}_8$ solution and the remaining PVP/PMF on graphene layer was inversely transferred onto heavily p-doped Si substrate.

Characterization of Doped Graphene Layer: The doped graphene on various PVP/PMF layers was investigated by PL/Raman spectroscopy (Alpha 300 M+, WITech). Raman spectroscope with an excitation wavelength of 532 nm was used, its layer beam size was about 0.7–0.9 μm , and the instrumental spectral resolution was below 0.9 cm^{-1} . An integration time of 5 s and spectrometer at 1800 grooves per mm were used for the test.

Fabrication and Electrical Measurements of Graphene Device: The transferred graphene was patterned by optical lithography and oxygen plasma etching processes to define the active channel region. Electron-beam lithography was also used to form edge contact patterns (5, 1, and 0.12 μm) on graphene underneath the metal pad. The distance between edge patterns and the metal edge was fixed as 15 μm in edge 2 and 3 patterns to clearly observe the surface current. Edge 2, 3, and 4 patterns, respectively, have 5, 1, and 0.12 μm wide periodic lines/gaps. Finally, 5 nm Ti and 50 nm Au layers were deposited to form source/drain metal electrodes by electron-beam evaporator. The fabricated devices were analyzed through electrical measurements (I_D – V_G) by HP 4155A semiconductor parameter analyzer, where Dirac points were extracted. M–G contact resistance values were obtained by TLM measurements.

Fabrication and Optical Measurements of Graphene/Perovskite Photodetector: For the fabrication of a back-gated perovskite/graphene hybrid structure photodetector, a perovskite absorbing layer was formed by a spin-coating process on graphene photodetectors with 2D surface and edge metal contacts (edge 1, edge 2, and edge 3). The photodetectors were then analyzed by current–voltage (I_D – V_G) measurement using an HP 4155A semiconductor parameter analyzer under both dark and illuminated conditions. The light source was a dot laser with wavelengths of 520, 655, 785, and 850 nm, where an optical power was 6 mW cm^{-2} . To characterize the optoelectronic performance of perovskite/graphene photodetectors with various contact patterns (surface, edge 1, edge 2, and edge 3), photoresponsivity (R) and detectivity (D^*) were calculated from the I_D – V_G . R is $I_{\text{photo}}/P_{\text{light}}$ and D^* is $(RA^{1/2})/(2eI_{\text{dark}})^{1/2}$, where I_{photo} is the generated photocurrent ($I_{\text{Laser on}} - I_{\text{Laser off}}$), P_{light} is the total incident optical power, A is the effective area of the detector, e is the absolute value of an electron charge ($1.6 \times 10^{-19}\text{ C}$), and I_{dark} is the dark current.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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